

🌐 No Sensors, No Problem: Building Cloud-Scale IoT Monitoring with Simulated Telemetry Streams

• Event-Driven • Scalable • Serverless • Cloud-Native

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- Expensive IoT hardware limits experimentation
- Infrastructure dependency reduces accessibility
- Limited access to real telemetry environments
- Difficult for beginners to prototype scalable monitoring systems
- Cloud-native IoT learning remains hardware-centric

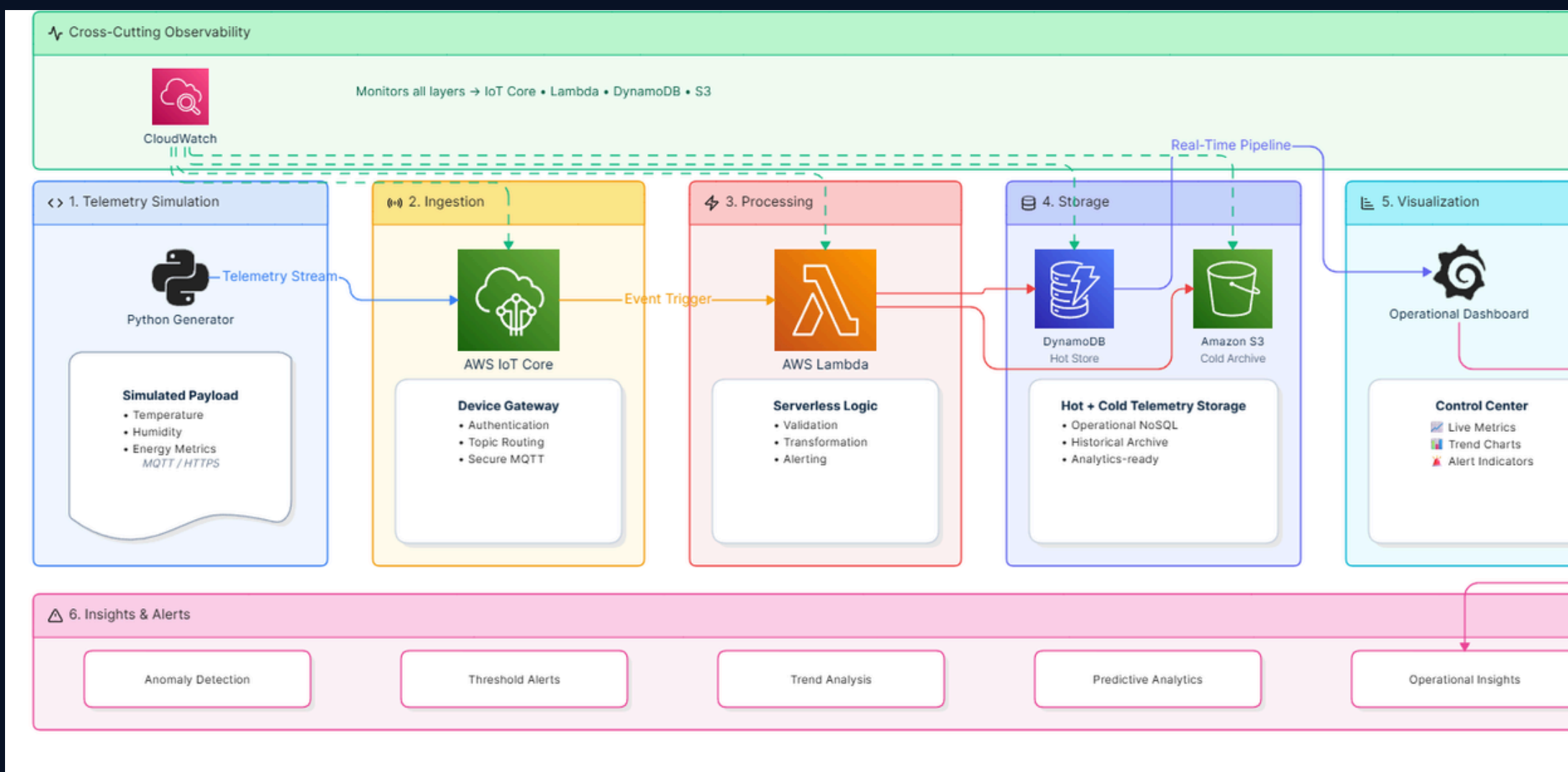
⚠️ Problem

💡 Solution



- Generates simulated real-time telemetry streams
- Implements scalable event-driven monitoring pipelines
- Demonstrates cloud-native observability workflows
- Enables hardware-independent IoT experimentation
- Supports accessible telemetry and monitoring education

Scalable event-driven telemetry ingestion and observability pipeline using cloud-native AWS services.



🚀 Impact

- ✓ Hardware-independent experimentation
- ✓ Beginner-friendly cloud-native learning
- ✓ Low-cost IoT prototyping



🔮 Future Scope

- Kubernetes deployment
- Edge telemetry integration
- AI anomaly detection
- Digital twin integration

📊 Monitoring Dashboard

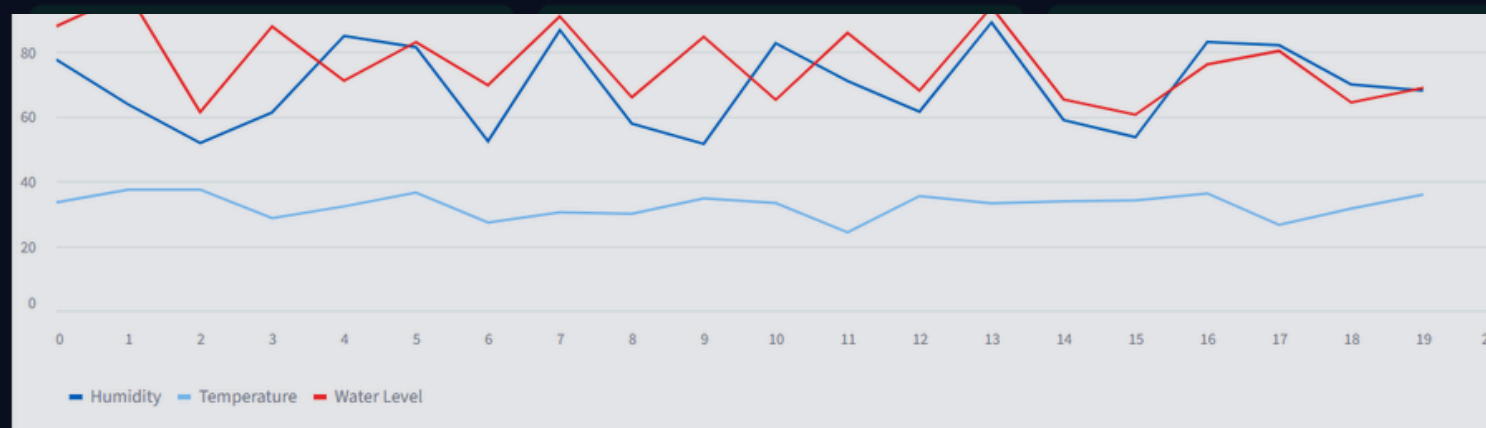
🌐 Cloud-Native IoT Monitoring Platform

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📈 Operational Metrics

Temp	Humidity	Water	Devices	Events/s	Latency
35.39°C	81.29%	93.57%	14	1199	1.41s

🟢 System Status



High Temperature Spike

Telemetry Delay Detected

Pipeline Monitoring Active

👁️ Observability

Logs Processed

14393

Streams

172

Success Rate

99.8%

→ OBSERVABILITY FEATURES →



CloudWatch

Monitors resources, collects metrics, and provides real-time insights.



Telemetry

Simulated IoT telemetry streams generated in real-time and processed at scale.



Monitoring

Tracks system health, visualizes metrics, and enables proactive alerting.

Beyond Hardware.
No Sensors. No Limits.